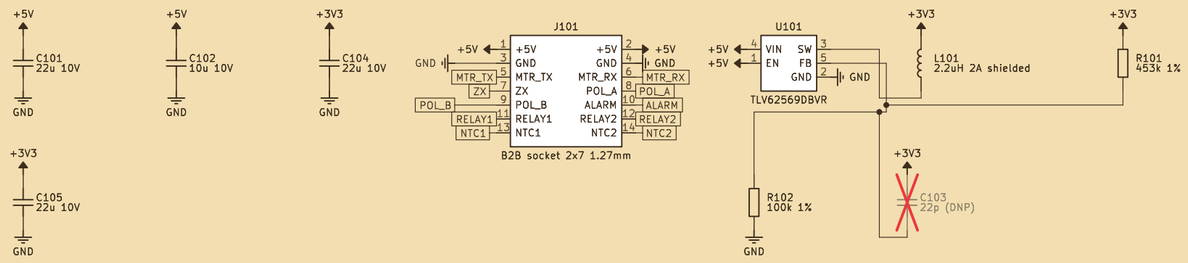
POWER AND PROCESSOR

3.3 V buck, ESP32-C6, programming pads.

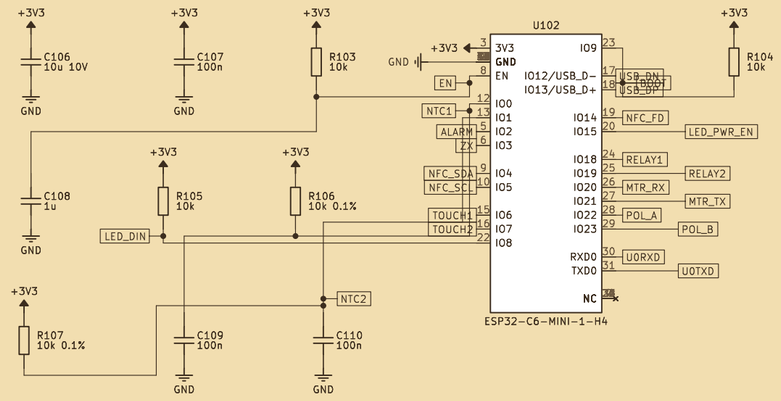
1. BOARD-TO-BOARD SOCKET AND 3.3 V BUCK

TLV62569: Vout = 0.6 V x (1 + 453k/100k) = 3.32 V. PFM at light load keeps standby low.



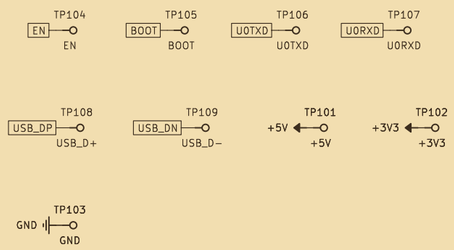
2. ESP32-C6-MINI-1-H4

Antenna at the faceplate rim, outside the steel box; keep-out per Espressif. Strapping: IO9 pulled up (SPI boot), IO8 pulled up.
ADC1: IO0 = NTC1, IO1 = NTC2. UART1: IO20 RX / IO21 TX to the BL0939. LEDC: IO18/IO19 relays. RMT: IO8 light lines.



3. PROGRAMMING AND TEST PADS

USB-Serial-JTAG on IO12/IO13 for flashing and logs; UART0 as fallback. Hold BOOT low at reset for download mode.



BOARD NOTES

FRONT BOARD, EVT1 DRAFT FOR REVIEW
Entirely SELV. 4 layers for the ESP32's RF ground; antenna keep-out at the board edge under the faceplate rim.
Connects to the rear board's safe zone only, through J101 (2 x 7, 1.27 mm).

ERC power flags

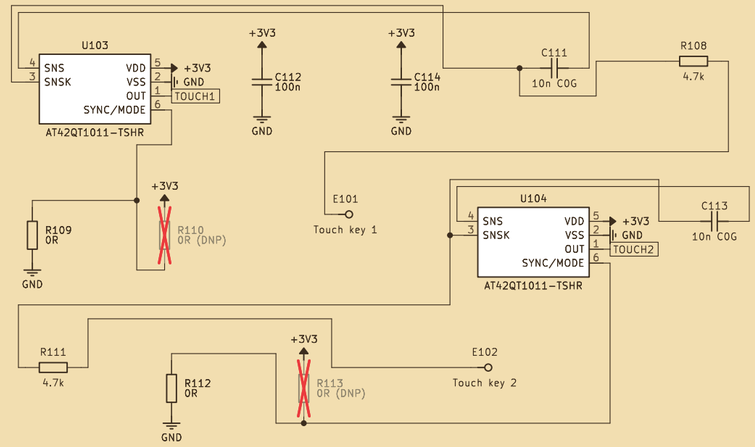


USER INTERFACE

Touch keys, NFC tag, light lines.

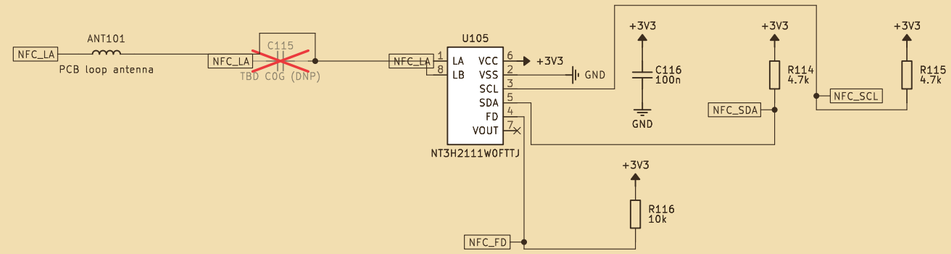
4. TOUCH KEYS (AT42QT1011), ONE PER OUTLET

Cs 10 nF and Rs 4.7k are starting values; tune on the EVT faceplate. OUT is active high: board_evt_c6 sets touch polarity.
SYNC/MODE strapped low by a OR (fit the other OR to strap high if the datasheet mode table calls for it).



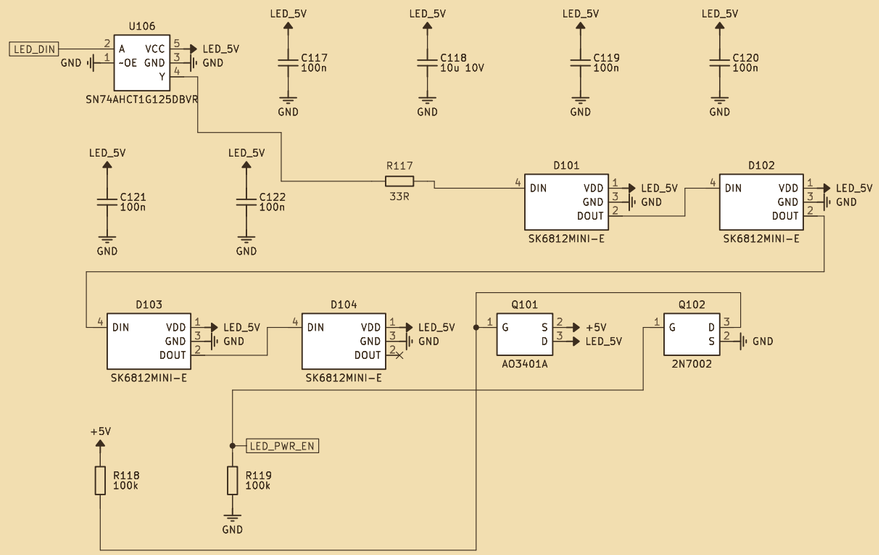
5. NFC TAG (NT3H2111)

Field-powered: a phone can read the setup record with the socket unpowered. Antenna tuned to 13.56 MHz with C115 at EVT.



6. LIGHT LINES (4 X SK6812MINI-E, TWO PER OUTLET)

LED power is switched off in standby (Q101); the buffer runs from the switched rail so nothing back-feeds the LEDs.
74AHCT inputs tolerate 3.3 V with VCC off. Data: IO8 -> buffer -> 33R -> D101 -> D102 -> D103 -> D104.



SELV only. Connects to the rear board's safe zone through J101.
Smart Wall Socket. DRAFT for engineering review, not a release.

Aether Labs

Sheet: /
File: sws-front.kicad_sch

Title: Smart Wall Socket EVT1, front board

Size: User	Date: 2026-11-10	**Rev: EVT1-draft-D**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1